

Exercises Cyber Security

1) Decrypting Codes

Decrypt following Texts. What is the original Text? What is the Code?

A)

Ejf Dbftbs-Wfstdimüttfmvoh (bvdi bmt Dätbs-Dijggsf, Dätbs-Bmhpsjuinv, Dbftbs-Wfstidijfcvoh, Wfstidijfcidijggsf pefs bmt Fjogbdifs Dbftbs cfAfjdiofu) jtu fjo fjogbdift tznfnusjdift Wfstidimüttfmvohtwfsbisfo, ebt bvg efs npohsbqijtdifo voe npopbmqibcfujtdifo Tvctujuvujpo cbtjfsu. Bmt fjoft efs fjogbdituf voe votjdifstuf Wfsbisfo ejfou ft ifvuf ibvqutädimjdi ebAv, HsvoeqsjoAjqjfo efs Lszqumphjf botdibvmjdi ebsAvtufmmfo. Efs Fjogbdiifju ibmcfs xfsefo pgunbmt ovs ejf 26 Cvditubcfo eft mbufjojtifo Bmqibcfut piof Voufstidijevoh wpo Hspß- voe Lmfjocvditubcfo bmt Bmqibcfu güs Lmbsufyu voe Hfifjnufyu wfsxfoefu voe TpoefsAfjdifo, TbuAAfjdifo vtx. ojdiu cfbdiufu.

B)

Af UjqhlgYjShZq, WfUjqhlagf ak IZW hjgUWkk gX WfUgVafY afXgjeSlagf. LZak hjgUWkk UgnWjlk IZW gjaYafSd jWhjWkWfSlagf gX IZW afXgjeSlagf, cfgof Sk hdSafIWpl, aflg Sf SdlWjSlanW Xgje cfgof Sk UahZWjIWpl. AVWSddq, gfdq SmlZgjarWV hSjlaWk USf VWUahZWj S UahZWjIWpl TSUc lg hdSafIWpl SfV SUUWkk IZW gjaYafSd afXgjeSlagf. wfUjqhlagf VgWk fgl alkWdX hjWnWfl aflWjXWjWfUW Tml VWfaWk IZW aflWddaYaTdW UgflWfl lg S ogmdV-TW aflWjUWhlgj.

xgj IWUZfaUSd jWSkgfk, Sf WfUjqhlagf kUZWeW mkmSddq mkWk S hkWmVg-jSfVge WfUjqhlagf cWq YWfWjSIWV Tq Sf SdYgjalZe. Al ak hgkkaTdW lg VWUjqhl IZW eWkkSYW oalZgml hgkKWkafY IZW cWq Tml, Xgj S oWdd-VWkaYfWV WfUjqhlagf kUZWeW, UgfkaVWjSTdW UgehmlSlagfSd jWkgmjUWk SfV kcaddk SjW jWimajWV. sf SmlZgjarWV jWUahaWfl USf WSkadq VWUjqhl IZW eWkkSYW oalZ IZW cWq hjgnaVWV Tq IZW gjaYafSlg lg jWUahaWflk Tml fgl lg mfSmlZgjarWV mkWjk.

zaklgjaUSddq, nSjagmk Xgje gX WfUjqhlagf ZSnW TWWf mkWV lg SaV af UjqhlgYjShZq. wSjdq WfUjqhlagf IWUZfaimWk oWjW gXIWf mkWV af eadalSjq eWkkSYafY. KafUW IZWf, fWo IWUZfaimWk ZSnW WeWjYWV SfV TWUgeW UgeeghdSUW af Sdd SjWSk gX egVWjf UgehmlafY. EgVWjf WfUjqhlagf kUZWeWk mkW IZW UgfUWhlk gX hmTdaU-cWq SfV kqeeWljaU-cWq. EgVWjf WfUjqhlagf IWUZfaimWk WfkmjW kWUmjalq TWUSmkW egVWjf UgehmlWjk SjW afWXXaUaWfl SI UjSUcafY IZW WfUjqhlagf.

C)

Jdlr Jlxolr Fhvdh (lq odwlqr: Jdlxv Lxolxv Fdhvdu, Surqxqfld uhvwdxudwd r fodvlfld; DIL: ['gä:mɔv 'mx:.ol.ɔv 'nɛ̃.vä:u]; qhooh hsljudil F·LYOLYV·F·I·FDHVDU h GLYYV LYOLYV; lq juhfr dqwlfr: Γάιος'Ιούλιος Καῖσαρ, Jáirv Lúolrv Ndîvdu; Urpd, 13 oxjolr 101 d.F. r 12 oxjolr 100 d.F. – Urpd, 15 pduCr 44 d.F.) è vwdwr xq plolwdh, srolwlfr, frqvroh, glwwdwruh, srqwhilfh pdvvlpr, rudwruh h vfulwwruh urpdqr, frqvlghudwr xqr ghl shuvrqdjil slù lpsruwdqwl h lqioxhqw ghoo vwruld.

CZZc sl psmjm dmlbYkclrYjc lcjjY rpYlqgxgmlc bcj qgqrckY bg emtcplm bYjjY dmpkY pcnsZZjgaYlY Y oscjjY gkncpgYjc. Ds bgrrYrmc (bgarYrmc) bg PmkY YjjY dglc bcj 49 Y.A., lcj 47 Y.A., lcj 46 Y.A. aml aYpgaY bcacllYjc c bYj 44 Y.A. amkc bgrrYrmc ncpncrsm, c ncp oscqrm pgrclsrn bY Qtcrmlgm gj npgkm bcg bmbgag AcqYpg, gl qcesgrm qglmlgkm bg gkncpYrmc pmkYlm. Aml jY amlosgrY bcjjY EYjjgY qcrcq gj bmkglgm bcjjY pcq nsZjgaY pmkYlY dglm Yjj'macYlm yrjYlrgam c Yj Pclm; nmprò ejg cqcpagr pmkYlg Y gltYbcpc ncp jY npgkY tmjrY jY zpgrYllgY c jY EcpkYlgY c Y amkZYrrcpc gl QnYelY, EpcagY, Cegrrm, Nmlrm c ydpgaY.

Lo sulpr wulxpyludwr, o'dffruger sulydwr shu od vsduwlClrqh gho srwhuh frq Jqhr Srpsr Pdjqh h Pdufr Olflqlr Fudvvr, vjhqò o'lqlClr ghoo vxd dvfhvd. Grsr od pruw h gl Fudvvr frqwr lo srsor ghl Sduwl, (Fduh, 53 d.F.), Fhvdh vl vfrquò frq Srpsr h od idClrqh ghjol rswlpdwhv shu lo frqwuroor ghoor Vwdwr. Qho 49 d.F., gl ulwruqr gdood Jdool, jxlgo oh vxh ohjlrql dwwudyhuvr lo Uxelfrqh (lq fxl Oxflr Fruqholr Vlood vwdelò lo qxrr frqilq do qrug gho srphulr ghoo flwwà qhoò'81 d.F.) qho surqxqfldqgr oh fhoheul sdurh «Dohd ldfwd hvw», h vfdwhqò od jxhuud flyloh, frq od txdo glyhqh fdsr lqglvfxvvr gl Urpd: vfrqilvh Srpsr d lduvdor (48 d.F.) h vxffhvvlydphqwh jol dowul rswlpdwhv, wud fxl Fdwraqh o'Xwlfhqvh, lq Diulfd h lq Vsdjqd.

Aml j'Yqqslxgmlc bcjjY bgrrYrspY Y tgrY bgcbc glgxgm Y sl nmpacqm bg pYbgaYjc pgdmpkY bcjjY qmagrà c bcjjY nmjgrgaY pmkYlY, Yqgaspylbgm nmrcpc Yqqmjsrm qsjjY PcnsZZjgaY. Gj qsm mncpYrm npmtmaò jY pcYxgmlc bcg amlqcptYrmc, dglafé sl epsnm bg qclYrmc, aYnceegYrg bY KYpam Egslgm zpsrm, EYgm AYqqgm Jmlelgl c Bcagkm zpsrm, amqngpò amlrpm bg jsg, saagbclbmjm Yjjc bgb bg kYpxm bcj 44 Y.A. (15 kYpxm 44). Lcj 42 Y.A., Ynncly bsc Yllg bmm gj qsm YqqYqqglgm, gj QclYrm jm bcdgaò sddgagYjklrc, cjctYlbgm Y bgtglrà. J'cpcbgrà pgdmpkYrpgac c qrmcgaY bg AcqYpc ds osglbg pYaamjrY bY MrrYtgYlm ysesqm, qsm npmlgnmrc c dgejgm Ybmrrgtm.

Oh fdpsdqh plolwdul h oh dClrqh srolwlfr h gl Fhvdh vrqr gd oxl vwhvvr ghwwdjoldwdphqwh udffrqwdwh lq whuCd shuvrqd qhl Frpphqwdull gh ehoor Jdoolfr h qhl Frpphqwdull gh ehoor flylol. Qxpurv qrwClh vxood vxd ylwd vrqr suvhqwl qhjol vfulwwl gl Dssldqr gl Dohvvdqgul, Vyhwrlr, Soxwdufr, Fdvvlr Glrqh h Vwuderq. Dowuh lqirupdClrqh srvrqr hvvhuh ulqudffldwh qhooh rshuh gl dxwrl vxrl frqwhpsrudhl, frph qhooh ohwwhuh h qhooh rudClrqh gho vxr ulydoh srolwlfr Flfhurq, qhooh srhvlh gl Fdwxor h qhjol vfulwwl vwrulfl gl Vdooxvwr.

D)

Vlqb (FxxpfeXrp: VlqX) mp hb llschpxXh zXr Mqemë ik lbx fbwqyvimgo gbrqvrq zXr hb vbkfs PXDfs ik hb Gfxqà Qbxosmsimqke hf VlqX GXtfxXpb (zlsolbik hb tossmkf Vlqb).

2) Code Definition

Define a two-way code for automated encryption and decryption that assures a good degree of security.

3) Security Analysis

Identify security threats and propose security measures for...

- A) ...a Smart Factory.
- B) ...a Data Warehouse.

- C) Define a systematic approach to security analysis of the former.

4) One-way Encryption

Describe the benefits and shortcomings of one-way encryption techniques.